

Theory in Practice: Testing the Limits of the Randles–Ševčík Equation

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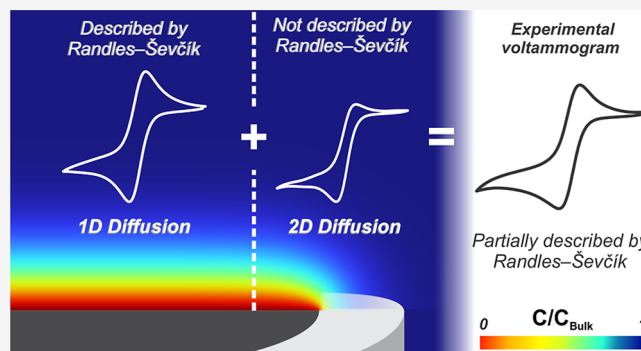
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ABSTRACT: The Randles–Ševčík equation remains one of the most used equations in electrochemistry, and when used within its boundary conditions, it is an analytical solution for a 1-dimension diffusion problem. Despite being derived under strict and very clear boundary conditions, experiments rarely observe them or work within these limits. One of the more elusive boundary conditions is that diffusion must be one-dimensional, which in theory would require an infinitely large, “edgeless”, electrode. Here, we systematically test the limits of the Randles–Ševčík equation across a wide experimental parameter space by combining cyclic voltammetry and numerical simulations. We use an electrically conductive thermoplastic filament to 3D print disk electrodes with radii ranging from 1 to 5 mm and use them to record cyclic voltammograms of $[\text{Ru}(\text{NH}_3)_6]^{3+}$ in aqueous solution at a wide range of scan rates. Experimental results reveal large positive deviations, exceeding 100%, from the Randles–Ševčík prediction for small radius electrodes and small scan rates, arising from contributions of radial diffusion and natural convection. Small negative deviations are seen for larger electrodes, and large scan rates are attributed to uncompensated Ohmic resistances. We propose the use of the integration of normalized voltammograms in the time domain (normalized charge) as a tool to quantify the experimental deviations from theory, establishing a practical framework for researchers to critically evaluate and quantify the applicability of the Randles–Ševčík equation in real electrochemical systems.

KEYWORDS: Randles–Ševčík equation, cyclic voltammetry, radial diffusion, natural convection, 3D-printed electrodes



INTRODUCTION

The Randles–Ševčík equation is ubiquitous in electrochemistry, describing the relation between the peak cathodic (or anodic) faradaic current in cyclic voltammetry with the square root of the voltammetric potential sweep rate (scan rate).^{1,2} The formulation of the equation came from polarography, derived from the current at mercury droplet electrodes and later extended to planar disc electrodes. The Randles–Ševčík equation (eq 1) is an analytical solution to a 1D diffusional problem, meaning that, when applied within its boundary conditions, it is exact and precisely predicts the faradaic current as a function of the scan rate (ν , V s^{-1}), geometric area of the disk electrode (A , cm^2), the diffusion coefficient of the species being redox cycled (D , $\text{cm}^2 \text{s}^{-1}$), its concentration (C , mol cm^{-3}), the number of electrons involved (n), and the temperature (T , K). F is Faraday's constant (C mol^{-1}) and R is the ideal gas constant ($\text{J mol}^{-1} \text{K}^{-1}$).

$$i_p = 0.4463nFAC \left(\frac{nF D \nu}{RT} \right)^{1/2} \quad (1)$$

When several voltammograms at different scan rates are recorded in the same solution, and the resulting peak faradaic currents are plotted vs the square root of the scan rate, a linear relation is expected, while working within the boundary

conditions, from which the diffusion coefficient, number of electrons, and concentration can be recovered from the slope of the linear fit. Beyond recovering these parameters, the plot of peak currents vs the square root of the scan rate is used to gauge the reversibility of the electrochemical process, said to be reversible, i.e., limited by diffusion, if the relation is linear.^{1–3} Conversely, if the linear relation is with the scan rate, not the square root of it, the process is said to be surface-limited, described by the Laviron equation.⁴ The use of the Randles–Ševčík equation to gauge the reversibility points to one of its boundary conditions that the process must be reversible. There are several other boundary conditions, all clearly stated in Randles' paper.¹

- The electrochemical process must be reversible, i.e., limited by diffusion, in Randles own words: “the concentrations of reactant and product can only be easily

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expressed in terms of the potential of the electrode [comment: Randles describes the potential sweep rate and the concentration as a function of time, which, when combined, results in the Nernst equation] if it is assumed that (1) the reaction product is soluble either in the aqueous solution or the material of the electrode, and (2) the electrode process is rapid compared with the rate of diffusion (i.e., the electrode reaction occurs reversibly)". The second point made by Randles makes it clear that the Randles–Ševčík equation describes a process that is kinetically reversible and, as such, the thermodynamic equilibrium between the oxidized and reduced species at the surface of the electrode (described by the Nernst equation) is maintained during the entire experimental time frame (here, the potential excursion). It is useful to recall that thermodynamics determines the position of equilibrium, and kinetics determines the rate at which equilibrium is approached. The Randles–Ševčík equation describes processes where the equilibrium position is reached very fast, almost instantaneously.

Another way to see this boundary condition is that the rate of mass transport (k_{mt}), must be slower than the rate of electron transfer reactions (k_0). In other words, k_{mt} must be much smaller than that of k_0 . This interplay is perhaps better described by Nicholson's dimensionless parameter (ψ), essentially the ratio of k_0 over k_{mt} .^{5,6} In Nicholson's work, processes with $\psi > 7$ are reversible, while ones with $\psi < 0.001$ are irreversible.⁵ This has since been revised several times, with some reports of $\psi > 100$ for reversible processes.⁷

- Mass transport is by diffusion only; no migration is considered due to the excess of supporting electrolyte. "Since the aqueous solution contains an excess of indifferent electrolyte, the activity coefficient of the reacting ions may be regarded as constant".¹ Convection is also not considered, although not directly expressed, the entire paper regards the diffusion problem.
- Only 1D semi-infinite diffusion is considered. "Although diffusion to a mercury drop or similar micro-electrode is not linear (. . .) equation applies sufficiently well (. . .) while the diffusion layer is thin in comparison with the diameter of the electrode" and "If the boundary concentrations (. . .) at the electrode surface and (. . .) in the bulk of the solution are known over a range of t (. . .), the concentration gradient close to the electrode surface and hence the diffusion current, for this range of t , are easily deduced".¹

The first boundary condition, the requirement for the electrochemical process to be reversible, is the reason why the electrode surface area, often recalled as the electrochemical surface area (ECSA), is not an adjustable parameter in the Randles–Ševčík equation.^{8,9} For a diffusion-limited process, the area to be considered, except on very short times, is always the geometric area, and not the surface area.⁸ The rationale is that if we are using diffusion as our "ruler", we cannot measure anything below the smallest graduation of said ruler, which is the mean diffusional length ($\langle x \rangle$), described by Einstein's equation (eq 2), where D is the diffusion coefficient and t is the time. Even at small times, the value of $\langle x \rangle$ will fall in the tens of micrometer ranges, well above the surface roughness of most electrodes, which usually are in the nanometer scale. In simple terms, diffusion quickly becomes blind to the electrode surface roughness.^{8,10,11}

$$\langle x \rangle = \sqrt{2Dt} \quad (2)$$

Although often neglected, the first boundary condition is better understood than the other two, which are rarely mentioned or considered. The second boundary condition, saying that mass transport is by diffusion only, is often acknowledged but not considered in its entire extension. Having an excess of supporting electrolyte is enough to mitigate migration contribution, but the absence of solution stirring or heating is not enough to avoid free convective contributions at short time scales.¹² Natural convection can greatly increase mass transport toward the electrode surface if cyclic voltammograms are performed at small scan rates.^{12–14} The third boundary condition, stating that diffusion must be one-dimensional (planar), requires the concentration gradient at the electrode surface to have 1D symmetry. This is only true for an infinitely large electrode, as the edge of an electrode will have radial diffusion, contributing to mass transport. If we consider all boundary conditions, particularly the last two, we can see that the Randles–Ševčík equation has a spatial and temporal limitation, and to find its limits, we must address what size of disc electrode and which scan rates result in the least deviations from the expected behavior from which the Randles–Ševčík equation was derived. Compton's group has already reported on the spatial limitation using numerical simulations,¹⁰ but here we set out to test these limits in practice, using 3D-printed disk electrodes¹⁵ of different radii. We systematically investigate the voltammetric profile of these electrodes in an aqueous solution of hexaammineruthenium chloride at different scan rates and use numerical simulations to rationalize our findings.

EXPERIMENTAL SECTION

Materials and Chemicals

All reagents were of analytical grade and used without further purification. Hexaammineruthenium chloride ($[\text{Ru}(\text{NH}_3)_6]\text{Cl}_3$) and potassium chloride (KCl) were purchased from Sigma-Aldrich. Conductive silver ink (RS PRO Silver Conductive Lacquer) was purchased from RS. 3D printing ABS-like resin was purchased from Elegoo, and 3D printing polylactic acid conductive thermoplastic filament containing carbon black was purchased from Protopasta. A commercial reference electrode (Ag/AgCl, KCl 3M from Metrohm) and a platinum counter electrode were used for all of the experiments.

Electrochemical Measurements

Cyclic voltammetry experiments were performed inside a Faraday cage using a PGSTAT 302N potentiostat from Metrohm, controlled by 2.1 NOVA software. All experiments were performed at a room temperature of 22 ± 0.5 °C. Before experiments, the contact resistance of the 3D-printed electrodes was measured and compensated in every cyclic voltammogram experiment by running a positive-feedback procedure within the NOVA program, and detailed in the equipment manual.¹⁶ Ohmic drop compensation is not limited to Metrohm Autolab instruments and can also be performed with other potentiostats.¹⁷ All voltammograms of the 3D-printed electrodes were recorded in a 1 mM $[\text{Ru}(\text{NH}_3)_6]\text{Cl}_3$ in 0.1 M KCl solution, over a potential window between 0.25 and -0.55 V vs the reference electrode. Voltammetric scan rates used were 1, 5, 10, 25, 50, 100, 150, and 200 mV s⁻¹, a range commonly found in the literature since larger scan rates (>200 mV s⁻¹) are typically avoided to minimize Ohmic drop effects, as these studies generally do not employ Ohmic drop compensation strategies.^{18–22} All experimental faradaic peak currents reported were isolated by baseline subtraction on the forward scan (cathodic peak) branch of the cyclic voltammogram.

Numerical Simulations

Numerical simulations of the diffusion layer profiles and cyclic voltammograms for the different electrode sizes were performed in



Figure 1. Manufactured 3D-printed electrodes ordered by radius (1, 1.5, 2, 2.5, 3, and 5 mm). Numbers printed on the electrode bodies are the electrode diameters.

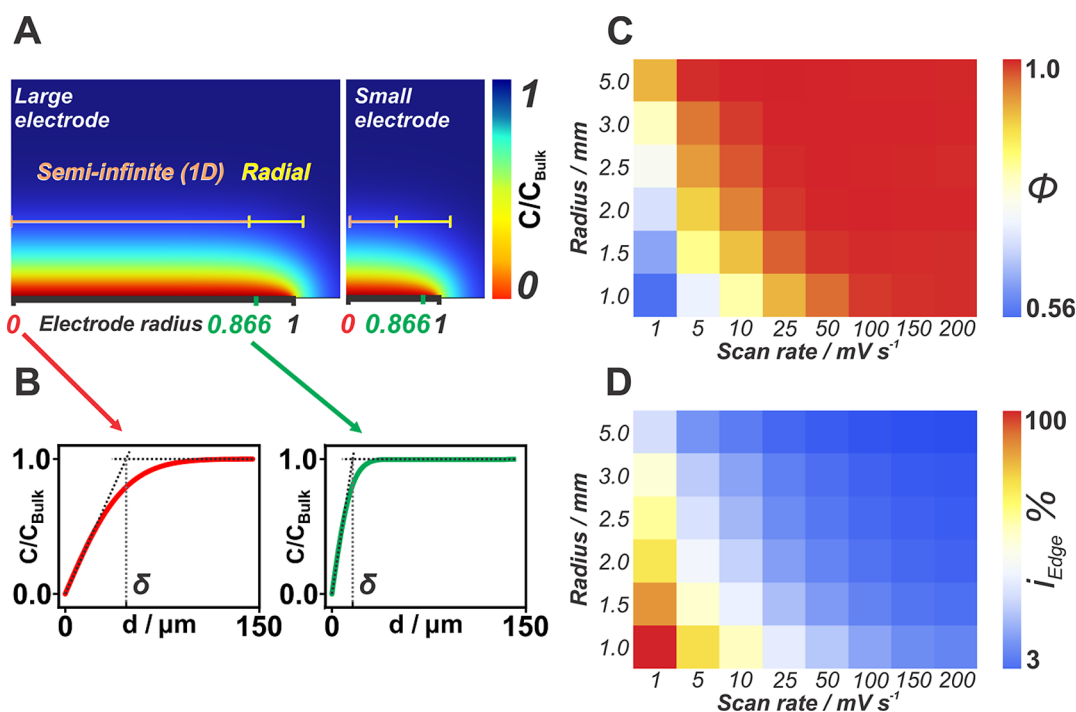


Figure 2. (A) Illustrative simulations of the diffusion layers at the most cathodic potential of the voltammogram for a large (5 mm radius) and a small (2 mm radius) disk electrode, pointing to the contributions of 1D and radial diffusion on mass transport. (B) Concentration profiles from radial coordinates 0 (in red) and 0.866 (in green) from which the diffusion layer thickness (δ) can be calculated. (C) Heatmap of the curvature of the diffusion layer (Φ) for the different electrode radii and voltammetric scan rates. (D) Heatmap of the contribution of the edge of the electrode (radial diffusion) to the overall current flowing at the electrode.

COMSOL Multiphysics v6.2, using the transport of the diluted species module. Details of the simulation can be found in Section 1, [Supporting Information](#).

Electrode Fabrication

Disk electrodes with 1, 1.5, 2, 2.5, 3, and 5 mm radii were fabricated using two different 3D printing technologies, masked stereolithography (MSLA – “resin printing”) and fused deposition modeling (FDM – “filament printing”).^{15,23} MSLA was used for fabricating the electrode housing, a 3 cm long, 1 cm radius cylinder containing a 5 mm deep cavity in its center, with the radius of the desired electrode. A 1 mm

diameter tube, centered on the cylinder, runs along the entire cylinder length, and it was used to house a copper wire forming the contact of the conductive disk with the potentiostat. The conductive disk was fabricated with a variation of the FDM printing method using a 3D printing pen to deposit the conductive thermoplastic filament in the corresponding cavity. Importantly, to mitigate contact resistance, which can be high in this type of electrode,^{15,24} the copper contact wire was first placed in the electrode, set in the bottom of the electrode cavity and secured with conductive silver ink prior to its filling with the conductive thermoplastic. Even though baseline subtraction was performed to isolate the Faradaic currents (vide supra), the specific capacitance of

3D-printed electrodes is rather small, and capacitive current contributions are minimized.²⁵ Before experiments, all electrodes were polished in increasing grit sandpaper, followed by progressively finer alumina suspensions until a mirror finish on the electrode surface was seen (Figure 1). Due to the manual fabrication method, all electrode sizes were fabricated in quadruplicate ensuring that at least three of each size worked and could be used in the experiments. It is worth noting that the smallest electrodes had the least reproducible fabrication and the largest contact resistance. Contact resistances were measured to be 390 ± 49 , 248 ± 45 , 174 ± 16 , 129 ± 18 , 98 ± 11 , and $57 \pm 5 \Omega$ for the 1, 1.5, 2, 2.5, 3, and 5 mm radius electrodes, respectively. All of the compensated resistance values are presented in Section 2, Supporting Information.

RESULTS AND DISCUSSION

Diffusion Layer Curvature

Before analyzing the experimental results, we must first consider the impact the different experimental conditions (electrode radius and scan rate) will have on the diffusion profile over the electrode surfaces, as this is key for understanding which conditions will satisfy the 1D diffusion boundary condition. A diffusion layer that is mostly planar will meet this requirement, but all diffusion layers have an edge, with more curved diffusion layers having more influence from radial diffusion. The curvature of the diffusion layer is not readily available from experiments, but it can be estimated from simulations. At one extreme, for the most planar layer, the diffusion layer will be mostly flat, with little contribution from the edges (Figure 2A). At the other, for the highest curvature, the diffusion layer will be a sphere cap, like in a microelectrode (Figure S2).²⁶

To quantify the curvature, simulations of all electrode sizes at the selected scan rates were performed, and the thickness of the diffusion layer (δ) at the most cathodic potential of the simulated voltammograms was calculated at two locations over the electrode surface, one at the center with a radial coordinate 0 and another near the edge, with a radial coordinate of 0.866 (Figure 2B). This coordinate value follows from the geometric properties of a circle; if the diffusion layer was a spherical cap, its height at this radial coordinate is expected to be half of the value of the electrode radius (Figure S2). Applying these properties, we can use the ratio between these two thicknesses to calculate the diffusion layer curvature (ϕ). If the ratio is 0.5, the diffusion layer is completely spherical, and if the value is close to 1, it is planar. Details on the curvature calculation can be found in Section 3, Supporting Information. These values can be used to construct a heatmap (Figure 2C) of ϕ for the different voltammetric scan rates and electrode radii. As expected, at slow scan rates, and particularly for small electrode radius, ϕ is close to 0.5, as the diffusion layer is very curved, while larger electrode radii and faster scan rates result in ϕ closer to 1 as the diffusion layer is largely flat. As curvature increases (i.e., ϕ decreases), radial diffusion contributes more to mass transport, violating Randles–Ševčík boundary conditions. Using simulations, we can quantify how much of the overall faradaic current is carried by the curved edges of the diffusion layer (Figure 2D), which should result in a surplus of current when compared to what is expected from 1D diffusion, as radial diffusion is more efficient.

Voltammetric Response

The limits of the Randles–Ševčík equation have been explored with simulations before,¹⁰ but to truly test it, experiments are important, as simulations usually only capture a few (but crucial) aspects of the experiments and might not be able to represent the

entire experiment. For instance, Compton's theoretical work focused only on diffusion, and as we will see, there are a few other aspects impacting electrodes in experiments that compound into larger deviations from the expected theoretical behavior. For testing the limits experimentally, cyclic voltammograms were recorded in 1 mM $[\text{Ru}(\text{NH}_3)_6]\text{Cl}_3$ in 0.1 M KCl solution with the electrodes fabricated with 6 different radii, at 8 different voltammetric scan rates (see above). Three electrodes of each radius were used for the experiments and data analyses. Figure 3A shows the voltammograms for 1, 2.5, and 5 mm radius

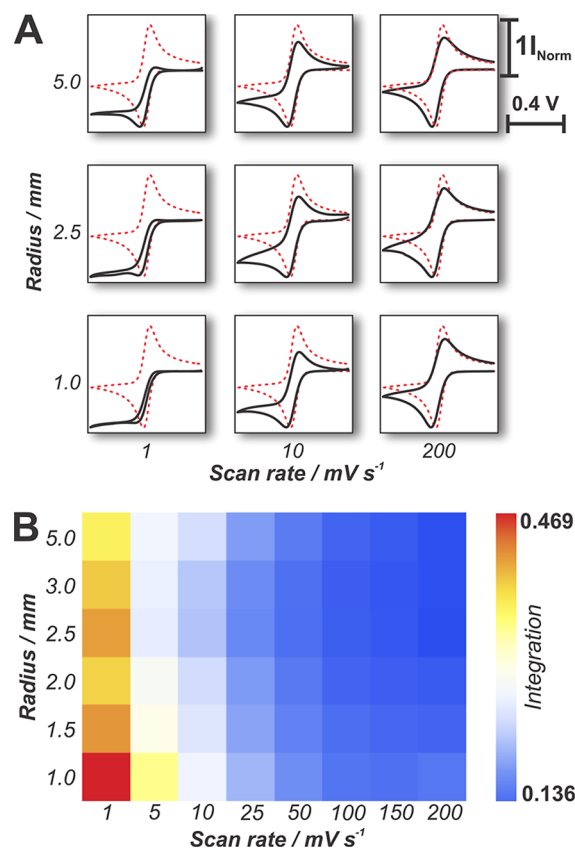


Figure 3. (A) Current normalized mean cyclic voltammograms ($n = 3$) recorded with the 1, 2.5, and 5 mm radius disk electrodes at scan rates of 1, 10, and 200 mV s^{-1} . Dashed red line shows the expected 1D diffusion only voltammograms (as expected by the Randles–Ševčík equation). (B) Heatmap of the mean integration value ($n = 3$) for the different electrode radii and voltammetric scan rates.

electrodes at scan rates of 1, 10, and 200 mV s^{-1} . The entire set of voltammograms for all electrodes can be seen in Figure S3 (Section 4, Supporting Information). All experimental voltammetric profiles are very different than that expected for 1D diffusion,³ as can be seen by comparing to the red dashed line voltammograms, which are for simulated voltammograms considering only 1D diffusion. At slow scan rates, and particularly for the smaller electrode sizes, it resembles the sigmoid profile of a microelectrode.²⁶

The trend in deviation from the expected reversible profile follows the predicted trend in Figure 2C, with small electrodes and slow scan rates resulting in larger deviations. This qualitative assessment can be made quantitative by integrating each voltammogram in the time domain after normalization by the maximum cathodic current and the total time (Figure S4A,B), resulting in the normalized charge that has flown through the

system. Under the experimental conditions used (only the oxidized form of the hexaammineruthenium salt in solution and the potential window centered at $E_{1/2}$), we can expect a well-defined range for the normalized charge values, following the two extreme cases: purely 1D diffusion with an integration value of 0.105 (Figure S4C), and only radial diffusion with an integration value of 0.5. More details about the integration process are presented in Section 5, Supporting Information. Figure 3B shows the heatmap of the mean integration values for all electrode sizes and voltammetric scan rates. All of the normalized voltammograms for all of the electrode sizes and scan rates can be seen in Figure S5. The integration values used to construct Figure 3B can be found in Table S4.

The voltammograms at the slowest scan rate (1 mV s^{-1}) with the smallest electrode (1 mm radius), have a mean integration value of 0.4696 ± 0.0108 , close to what is expected for a sigmoid-shaped voltammogram (see Figure S4D). It is interesting that this is different from what is expected from simulations at this condition, considering diffusion only, which show voltammetric profiles with defined oxidation and reduction peaks (Figure S6A) and an expected integration of 0.222. This suggests that diffusion alone cannot explain these deviations in the experimental voltammograms. This is further confirmed when we look at the plot of the contribution of radial diffusion to the current carried by the electrode, derived from the data in Figure 2D, with the integration value (Figure S6B). Although there is a positive correlation between the contribution of radial diffusion and the integration value, the data are very dispersed, particularly at slow scan rates. At these slow scan rates, natural convection can strongly affect mass transport rates.¹² If we consider this effect in the simulations (Figure S6C),^{12–14} we can see that the simulated voltammogram matches the experimental much closely than when only diffusion was considered. This introduces the concept that the limitations in using the Randles–Ševčík equation are not only spatial but also temporal. At slow scan rates not only the diffusion layer grows more curved (Figure 2B), but also at these longer times natural convection contributes strongly to mass transport rates, resulting in voltammograms with high integration values, even for the 5 mm radius electrode (Figure 3B). At faster scan rates, natural convection is unimportant, and the integration values are close to the theoretically expected ones.

Quantifying the Error

We have shown and quantified the deviations of the experimental voltammograms from the 1D diffusion reversible voltammogram, but we have not yet quantified the error in using these electrode sizes and scan rates with the Randles–Ševčík equation. To calculate the error, we must consider the experimental faradic peak currents for the different electrodes and scan rates and compare them to those predicted by the equation, with the difference between these values being the error. Although this sounds easy, there are some challenges associated with experimental errors in the solution concentration value, temperature fluctuations and, more importantly, the precise measurement of the electrode radius, which compounds to larger errors when using the equation. 3D printers are precise but not exact,²³ and particularly for the smaller electrodes, the error is within an order of magnitude (100s of μm) from the radius size. The size of the electrodes also falls in an unfortunate range where they are mostly too large for electron and optical (for the latter, field of view is the limiting aspect) microscopy characterization and too small for measuring

with calipers precisely. As this is further compounded by the Randles–Ševčík equation, considering area and not radius, we took a different approach for calculating the error.

There are two aspects to consider in our error calculation approach. First, we must understand that any linearization of the Randles–Ševčík equation must intercept the y axis (faradaic peak current value) at zero when the x axis (square root of the scan rate) is zero. Second, supported by our simulations and integration values (Figures 2C and 3B, respectively), deviations from Randles–Ševčík in our data set are more significant on the first few scan rate values, with most of the other data points, which are at faster scan rates, having much smaller deviations. As there are more data points with small deviations than with large deviations, the smaller deviation data points will bear more influence in the linear regression. Considering that, for a given electrode size, the voltammograms at different scan rates were recorded in succession, the experimental conditions within the experimental data set of a single electrode did not change, and a simple linearization of the experimental data points, fixing the intercept to zero, must return a linear equation that is very close to the Randles–Ševčík expression for that experiment. We can then calculate the error for each experimental data point by measuring its distance from the Randles–Ševčík linearization. More details on the error calculation can be seen in Section 6, Supporting Information.

Figure 4A shows a heatmap built with the mean experimental error, which shows a similar trend observed throughout the

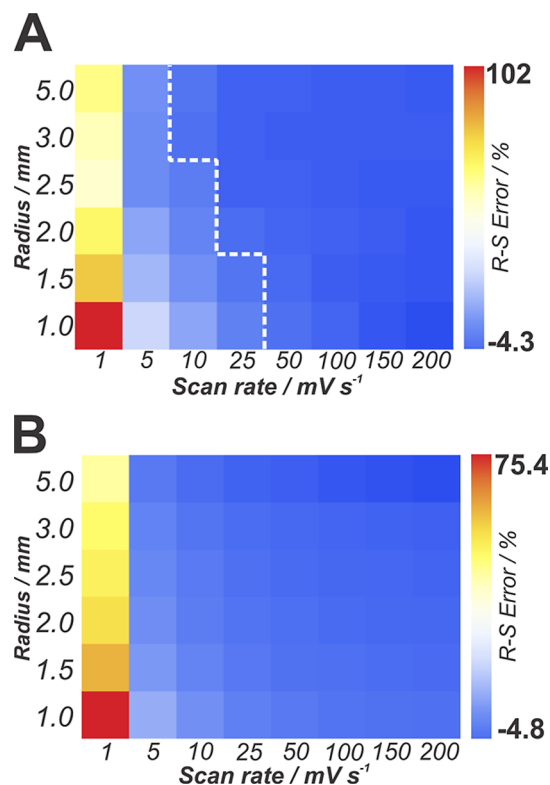


Figure 4. (A) Heatmap of the mean experimental error (experimental faradaic current distance from the expected by the Randles–Ševčík linearization— $n = 3$) for the different electrode radii and voltammetric scan rates. All values to the left of the dashed white line have an error of $>5\%$. (B) Heatmap of simulated error (simulated faradaic current distance from the expected by the Randles–Ševčík equation), when simulations consider natural convection and an uncompensated resistance of 10 Ohms .

manuscript. At slow scan rates, and particularly for small electrodes, a high positive error is seen, indicating an “excess” of faradaic current when compared to what is expected from the Randles–Ševčík equation. For the 1 mm radius electrode at a scan rate of 1 mV s^{-1} , this error is $102 \pm 23\%$. As the equation considers only 1D diffusion, the excess current results from the increased mass flow rate from the other mass transport modalities present, radial diffusion (see Figure 2C,D), and natural convection (see Figure S6C). At fast scan rates, a negative error is seen, up to $-4.30 \pm 0.54\%$ for the 1 mm radius electrode at a scan rate of 200 mV s^{-1} , indicating a lack of faradaic current when compared to that expected from the Randles–Ševčík equation. This “missing” current results from a system inefficiency in the form of Ohmic losses, more pronounced at the larger current values and, in our experiments, with the more resistive electrodes (1 mm radius, Table S3). This trend is clearly illustrated in Figure S7. Even with the least resistive electrodes (5 mm radius—Table S3), at a scan rate of 200 mV s^{-1} , we find an error of $-0.81 \pm 0.22\%$. Although the experiments were performed with Ohmic drop compensation, this cannot be done to 100% of the resistance value, and the residual resistance manifests as the negative errors. This is particularly important for 3D-printed electrodes, which are notorious for their contact resistance and should be a lesser concern for metallic, commercial, and other electrodes. It should be noted that most commercial disk electrodes have radii between 1 and 2 mm, and often slow scan rates are employed for experiments involving the Randles–Ševčík equation. These conditions result in the largest experimental errors, which make the comparisons of experimental values with theoretical ones troublesome.^{18–22} In Figure 4A, all values to the left of the white dashed line have errors $>5\%$, and highlights how experimentally, the use of the Randles–Ševčík equation can carry heavy errors.

As we touched on before, truly investigating the limits of the Randles–Ševčík equation requires experimental data, as simulations often do not represent every single aspect of the experimental condition. This is showcased by the heatmap in Figure 4A, which shows errors that are not predicted by models, which account for diffusion only.¹⁰ The experimental aspects involving the use of the Randles–Ševčík equation are much more complicated, and natural convection and uncompensated resistances must be considered. If we consider only diffusion, our simulations naturally agree with Compton’s (Figure S9), and we find an error of 0.7% for the 5 mm radius electrode at a scan rate of 200 mV s^{-1} , similar to the 0.74% error for an 8 mm radius electrode reported.¹⁰ This contrasts with experiments (Figure 4A), where a negative error of $-0.81 \pm 0.22\%$ is seen for the same condition. Also, when we consider diffusion only, the largest error seen in the simulation is 47% (Figure S9) for the 1 mm radius electrode at a scan rate of 1 mV s^{-1} , which is much less than the $102 \pm 23\%$ seen in experiments (Figure 4A). If we consider natural convection and a 10 Ohm uncompensated resistance in our simulations (Figure 4B), the calculation errors range from -4.8% to 75.4% , much closer to the experimental values found. This shows that the experimental errors associated with the use of the Randles–Ševčík equation are indeed multifactorial and not only due to the contribution of radial diffusion to mass transport.

Knowing the Experimental Error

As the experimental error heatmap demonstrates (Figure 4A), there will always be an error when comparing experimental faradaic currents to the expected theoretical currents from the

Randles–Ševčík equation. This arises from contributions by radial diffusion, natural convection, and uncompensated resistances to the voltammetric profile, with each affecting a particular region on the experimental parameter space (electrode size and scan rate), as we have discussed. The region of the parameter space that is affected and the extension of the effect that each of these unaccounted aspects has on the voltammetric profile are dependent on the chemical environment. Particularly, changes in solution viscosity and diffusion coefficient will shift the effects within the parameter space, and the heatmap in Figure 4A is not universal. Thus, setting a hard limit to the Randles–Ševčík equation is impossible as it depends on many experimental aspects. A better alternative is to provide means for quickly measuring the error from experimental data, allowing researchers to know the errors associated with their experimental data and decide if these are reasonable or to move the experimental parameters to a region where error is deemed acceptable. Although Figure 4A serves as a guide for making this decision in aqueous solutions, it is derived from our experimental data and experimental conditions. As some of the error originates from uncompensated resistance, which can be large for 3D-printed electrodes,¹⁵ the error at larger scan rates might change for different sets of electrodes and different Ohmic drop compensation parameters. The lower left corner of the heatmap (Figure 4A), with the errors dominated by natural convection and radial diffusion, should be consistent for most experiments in 0.1 M KCl aqueous solutions, with a small variation due to temperature changes.

Constructing a bespoke heatmap for every single experimental data is impractical and unnecessary. The area under the curve (integral) of normalized voltammograms can be used to measure the deviations in experimental faradaic currents from the expected ones from the Randles–Ševčík equation. The integration value can quantify the shape of the voltammogram between a reversible, “Nicholson and Shain”, voltammogram and a sigmoid, microelectrode-like one. The deviations on the voltammetric profile arising from radial diffusion and natural convection, and the resulting area under the curve, are captured by the integration value, so it can be used as a tool to quantify how far the experimental currents are from theory. The impact of uncompensated resistances is hardly captured by the integration value since its impact on the voltammetric profile should be symmetrical. This is not an issue since most experiments are performed in high electrolyte concentration and, when performed correctly,^{15,27} uncompensated resistances are minimal, and so is their influence on the voltammetric profile. Even when we consider a large uncompensated resistance of $10 \text{ } \Omega$, the resulting error is approximately -5% , which is much smaller than the errors from radial diffusion and natural convection (Figure 4A,B).

Figure 5 shows the plot of the integration value vs the calculated error (deviation from Randles–Ševčík linearization), using the 3D-printed electrodes, and an exponential relation is clear (data in black). Data sets (integration values and error from Randles–Ševčík equation) recorded in a 1 mM $[\text{Ru}(\text{NH}_3)_6]\text{Cl}_3$ in 0.1 M KCl solutions with a 1.5 mm radius disk glassy carbon electrode (Figure 5, green data) and a 0.75 mm radius disk platinum electrode (Figure 5, blue data) are also presented in Figure 5. All of the voltammograms recorded with commercial electrodes are presented in Figure S10. As can be seen, the same exponential relation is observed, regardless of the electrode used, demonstrating that the integration value is a universal tool, within a given experimental condition (temperature, media

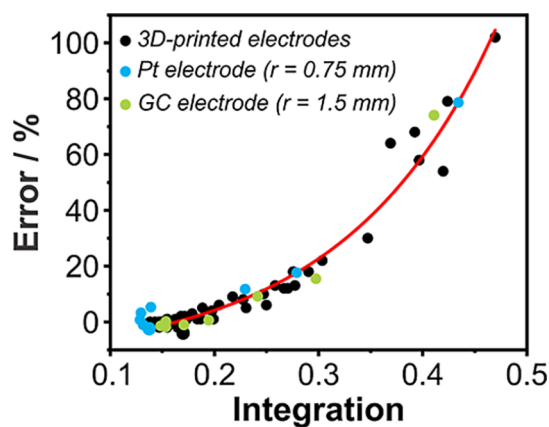


Figure 5. Plot of the voltammogram integration value for the different electrode radii and voltammetric scan rates vs the calculated error (deviation from the Randles–Ševčík linearization) showing an exponential relation (black dots). Red line is the exponential line of best fit. Also presented is the data for a 1.5 mm radius disk glassy carbon (GC) electrode (green dots) and a 0.75 mm radius disk platinum electrode (blue dots). Data are for 1 mM $[\text{Ru}(\text{NH}_3)_6]\text{Cl}_3$ in 0.1 M KCl aqueous solution, recorded at 22 °C.

composition, and viscosity), to quantify how far the experimental currents are from what is predicted by the Randles–Ševčík equation.

Although the exponential fit (see Section 7, [Supporting Information](#)) does not capture electrode kinetics ([Figure S11](#)) and the effect from uncompensated resistances, as discussed above and highlighted by the platinum electrode data at low integration values ([Figure 5](#), blue data), it can still be used as a semi-quantitative tool. Large integration values will result in larger deviations, while small integration values result in small errors. Integration values of less than 0.28 will result in deviations from the Randles–Ševčík equation of less than 20%, while integrations of less than 0.24 will lead to errors smaller than 10%. Further details of the error calculation from the integration values can be seen in Section 7, [Supporting Information](#).

CONCLUSIONS

We have systematically tested the practical limits of the Randles–Ševčík equation in 1 mM $[\text{Ru}(\text{NH}_3)_6]\text{Cl}_3$ in 0.1 M KCl aqueous solutions by combining numerical simulations and experiments using 3D-printed disk electrodes of different radii and performing cyclic voltammetry over a wide range of scan rates. We demonstrate that the deviations from the expected faradaic currents come from a combination of factors, namely, radial diffusion, natural convection, and uncompensated resistance, which are not considered in the Randles–Ševčík boundary conditions. These effects manifest differently across the experimental parameter space, with radial diffusion and natural convection contributing to larger experimental currents than expected (more than 100% error) at slow scan rates and for small electrode radii. Ohmic losses from uncompensated resistance become more significant at fast scan rates and larger electrodes, contributing to smaller experimental currents than expected. We reason that the experimental applications of the Randles–Ševčík equation will always be associated with errors, which can be minimized by working with large electrodes at fast scan rates (being mindful of the increased capacitive contributions at these conditions). Regardless, when invoking

the Randles–Ševčík equation, the error must be known. Rather than establishing a limit where the Randles–Ševčík equation can be used, we propose a simple tool for quantifying the experimental error, the integral of normalized voltammograms (normalized charge), which we demonstrate to correlate with the experimental faradaic current error. Our proposed metric captures the contributions of diffusion (1D and radial) and natural convection on the voltammetric profile and works as a universal quantifier for experimental error. By demonstrating the practical aspects limiting the application of the Randles–Ševčík equation, and providing a practical tool to assess experimental errors, we hope to enable researchers with the theoretical framework and tools to use the Randles–Ševčík equation critically and, hopefully, more appropriately.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acselectrochem.5c00457>.

Containing detailed information on the numerical simulations, values of the contact resistance of the 3D-printed electrodes, details on the calculation of the diffusion layer thickness and curvature, integration of the normalized voltammograms, and calculation of the experimental error ([PDF](#))

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Notes

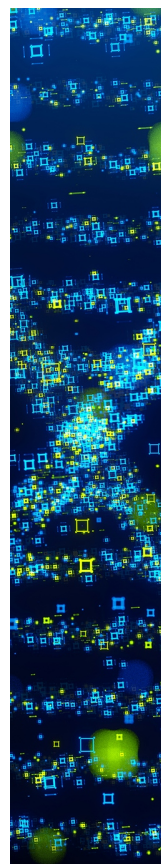
The authors declare no competing financial interest.

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